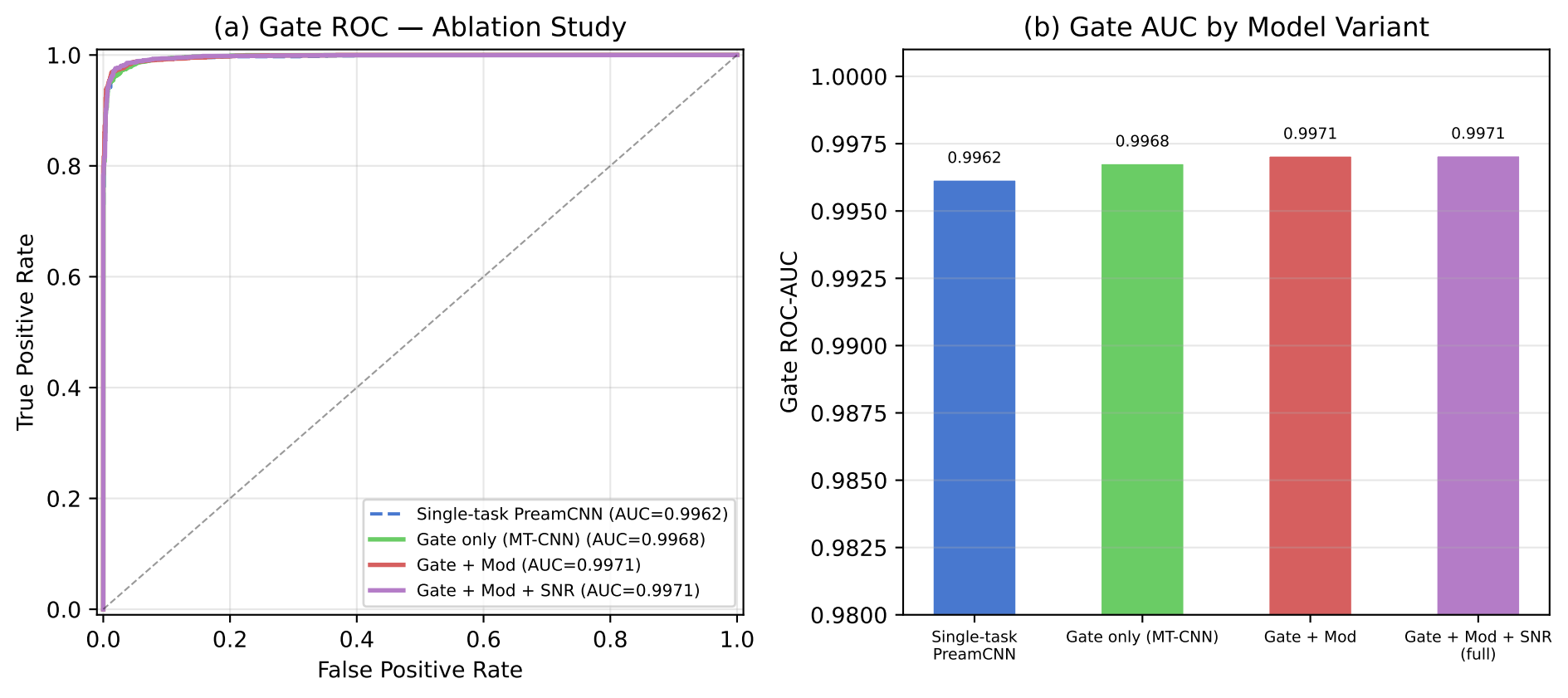
(a) Gate ROC — Ablation Study
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Single-task PreamCNN (AUC=0.9962)
Gate only (MT-CNN) (AUC=0.9968)
Gate + Mod (AUC=0.9971)
Gate + Mod + SNR (AUC=0.9971)
(b) Gate AUC by Model Variant
Gate ROC-AUC
1.0000
0.9975
0.9950
0.9925
0.9900
0.9875
0.9850
0.9825
0.9800
0.9962
0.9968
0.9971
0.9971
Single-task PreamCNN
Gate only (MT-CNN)
Gate + Mod
Gate + Mod + SNR (full)